

LEADED SOLDER PASTE

Complete Technical Guide

SnPb Alloys • Powder Types • Flux Classifications
Component Compatibility • Storage & Handling
Reflow Profiles • Quality Control

PCBA Manufacturing & Assembly Standards

IPC/J-STD-005 | IPC/J-STD-006 | IPC/J-STD-004 | RoHS Exemptions

1. OVERVIEW OF LEADED SOLDER PASTE

Leaded solder paste is a homogeneous mixture of **tin-lead (SnPb) solder alloy powder** and **flux medium**. For over 50 years, it was the universal standard for Surface Mount Technology (SMT) reflow soldering. Despite RoHS restrictions, SnPb paste remains critical in exempt applications where proven reliability, superior wetting, and lower process temperatures are mandatory.

Composition Breakdown:

- **Solder Powder (85–90% by weight):** Sn63Pb37, Sn62Pb36Ag2, or Sn60Pb40
- **Flux Medium (10–15% by weight):** Rosin, activators, solvents, rheology modifiers
- **Metal Load:** Typically 88–92% for standard applications; 85–87% for fine-pitch

Why Leaded Paste Still Matters:

- **Low melting point (183°C eutectic):** Reduces thermal stress on components and PCBs
- **Superior wetting:** Spreads faster and more uniformly than any lead-free alloy
- **Shiny joint appearance:** Bright, smooth fillets — easy visual inspection
- **Wide process window:** More forgiving reflow profiles, less voiding
- **Lower energy cost:** 30–40% less reflow oven energy consumption
- **Proven reliability:** Decades of field data in military, medical, and aerospace

■■ **Regulatory Status:** SnPb solder paste is **RESTRICTED** under RoHS Directive 2011/65/EU for most consumer electronics. It may only be used in exempt categories: medical devices, military/aerospace, monitoring instruments, and repair of pre-2006 equipment.

2. LEADED SOLDER PASTE ALLOY TYPES

2.1 Sn63Pb37 — Eutectic Tin-Lead

Sn63Pb37 is the most common leaded solder paste alloy. The 63% tin / 37% lead composition provides a **single, sharp melting point at 183°C** with no plastic range. This makes it ideal for all SMT processes where minimal thermal exposure and maximum wetting are desired.

Technical Specifications:

- **Melting Point:** 183°C (eutectic — liquidus = solidus)
- **Recommended Peak Reflow:** 210–230°C
- **Time Above Liquidus (TAL):** 30–60 seconds
- **Joint Appearance:** Bright, shiny, smooth surface
- **Wetting Speed:** Excellent — fastest of all common alloys
- **Voiding Tendency:** Very low — best-in-class
- **Shear Strength:** ~25 MPa (adequate for most applications)
- **Thermal Fatigue:** Good, but inferior to SAC305 in high-cycling applications

Best Applications:

- General-purpose SMT and through-hole paste-in-hole
- Medical devices (exempt from RoHS — pacemakers, implants)
- Military and aerospace electronics (MIL-STD-883 compliance)
- High-reliability industrial controls
- Repair and rework of legacy SnPb assemblies
- Prototype and low-volume production

■ **Process Advantage:** The low melting point allows faster conveyor speeds, shorter reflow profiles, and significantly lower energy consumption. Tip life in hand soldering is 2–3× longer than with SAC305.

2.2 Sn62Pb36Ag2 — Silver-Bearing Leaded

Sn62Pb36Ag2 (commonly called "SN62") contains 2% silver to prevent **silver scavenging** when soldering to silver-plated components or PCBs. Without silver, SnPb alloys dissolve silver pads and traces, creating brittle Sn-Ag intermetallics and weak, porous joints.

Technical Specifications:

- **Melting Point:** 179°C (slightly lower than Sn63Pb37)
- **Recommended Peak Reflow:** 210–225°C
- **Wetting:** Excellent — silver improves fluidity
- **Silver Scavenging:** Eliminated — safe for Ag-plated surfaces
- **Mechanical Strength:** Higher than Sn63Pb37 due to Ag content
- **Joint Appearance:** Bright and shiny
- **Cost:** Higher than standard SnPb due to silver content

Best Applications:

- Soldering to silver-plated PCBs or component leads
- High-reliability military and aerospace assemblies
- RF and microwave circuits with silver traces
- Switch and relay contacts with silver plating
- Hybrid microcircuits and power modules

■■ **Critical Warning:** Never use Sn63Pb37 paste on silver-plated surfaces. The lead dissolves silver rapidly, creating a brittle Sn-Ag intermetallic and leaving a weak, porous joint. Always specify Sn62Pb36Ag2 for silver-bearing substrates.

2.3 Sn60Pb40 — General Purpose

Sn60Pb40 is a near-eutectic alloy with a slightly wider melting range (183–190°C). The extra lead content makes it slightly softer and more ductile than Sn63Pb37. Historically the most economical choice when lead was cheaper than tin.

Technical Specifications:

- **Melting Range:** 183–190°C (non-eutectic, small plastic range)
- **Wetting:** Very good — nearly equal to Sn63Pb37
- **Joint Appearance:** Bright and shiny
- **Ductility:** Higher than Sn63Pb37 (more lead = softer alloy)
- **Cost:** Slightly lower than Sn63Pb37 historically

Best Applications:

- General consumer electronics (pre-RoHS legacy production)
- Cable and wire harness assembly
- Lower-cost through-hole paste-in-hole
- Repair of older equipment where exact alloy match is not critical

2.4 Other Leaded Paste Alloys

Sn50Pb50 Paste

Higher lead content (50%) creates a wider melting range (183–212°C) and a very soft, ductile alloy. Used for applications requiring maximum joint flexibility or where lower tin content reduces tin pest risk (allotropic transformation of tin at low temperatures).

- **Melting Range:** 183–212°C
- **Use:** Step-soldering, ductile joints, plumbing-adjacent applications

Sn40Pb60 & Sn30Pb70 Paste

Very high lead content alloys with melting points above 183°C (up to 275°C for Sn30Pb70). Used for **step-soldering** — soldering components at progressively lower temperatures so previously soldered joints don't remelt. Also used for non-electrical applications.

- **Melting Range:** 183–234°C (Sn40Pb60); 183–275°C (Sn30Pb70)
- **Use:** 2nd/3rd pass soldering, high-temp plumbing, step-sequence assembly

3. SOLDER POWDER PARTICLE SIZE (TYPES)

Leaded solder powder is classified by particle size distribution per **IPC/J-STD-005**. The type number determines stencil aperture design, printability, and minimum component pitch capability.

Type	Particle Size (μm)	Min Stencil Aperture	Best For
Type II	45–75	> 250μm	Coarse SMT, large pads, through-hole paste
Type III	25–45	200–250μm	Standard SMT, 0402/0603, SOIC, QFP
Type IV	20–38	150–200μm	Fine pitch QFP, 0201, small QFN
Type V	15–25	100–150μm	Ultra-fine 01005, micro-BGA, CSP
Type VI	5–15	< 100μm	Jet printing, 008004, advanced SiP
Type VII	2–11	< 80μm	Next-gen jet dispensing, wafer bumping

Table 1: IPC/J-STD-005 Solder Powder Classification (SnPb & Lead-Free)

Selection Guidelines:

- **Type III** is the workhorse for 95% of SMT applications
- **Type IV** required for 0201 passives and fine-pitch QFP (< 0.5mm pitch)
- **Type V/VI** used with jet dispensers (no stencil) for ultra-miniature components
- **Smaller particles = higher surface oxidation** → shorter stencil life, more voiding

■ **Storage Note:** Type V, VI, and VII powders have exponentially higher surface area and oxidize faster. These pastes require stricter refrigeration (0–5°C) and shorter shelf life (3 months vs 6 months for Type III).

4. FLUX CLASSIFICATIONS & CHEMISTRY

Flux is the chemical heart of solder paste. It cleans oxides, promotes wetting, and protects liquid solder from re-oxidation during reflow. Lead-based fluxes are formulated for lower temperatures (peak 210–230°C) and generally offer wider process windows than lead-free fluxes.

4.1 IPC Flux Activity Levels

Code	Activity	Halide Content	Residue	Use Case
L0	Low	0.0%	Non-corrosive, benign	No-clean, high-impedance circuits
L1	Low	< 0.5%	Non-corrosive, benign	No-clean, general SMT
M0	Moderate	0.0%	May be corrosive if not cleaned	Water-washable, high reliability
M1	Moderate	0.5–2.0%	Corrosive if not cleaned	RMA, water-wash, military
H0	High	0.0%	Corrosive — must clean	Difficult-to-solder surfaces
H1	High	> 2.0%	Corrosive — must clean	RA flux, plumbing (NOT PCBA)
RMA	Mildly Active	< 0.5%	Benign if low halide	Traditional electronics, military
RA	Fully Active	> 2.0%	Corrosive — must clean	Plumbing ONLY — never PCBA

Table 2: IPC/J-STD-004 Flux Classification for Electronics

4.2 Flux Chemistry Types for SnPb

RMA (Rosin Mildly Activated)

The traditional standard for SnPb electronics. Natural rosin (colophony) with mild activators. Safe for most applications. Residues are generally benign but can be cleaned if required. Widely used in military (MIL-STD) and medical exempt applications.

No-Clean (L0 / L1)

Modern no-clean fluxes for SnPb offer very low residue and are electrically safe. However, SnPb no-clean is less common than lead-free no-clean because RMA is so well-proven. Used when cleaning costs must be eliminated.

Water-Soluble (WS / M0-M1)

Strong activators provide excellent wetting. Residues are hygroscopic and corrosive — **must be washed** with DI water after reflow. Common in military, medical, and high-reliability applications where zero residue is mandatory.

RA (Rosin Activated) — ■■ NOT FOR PCBA

RA flux is too aggressive for printed circuit boards. It leaves corrosive residues that will damage traces and components over time. Strictly reserved for plumbing, stained glass, and non-electrical soldering.

5. COMPONENT TERMINATION COMPATIBILITY

SnPb solder paste must wet and bond to component terminations. Most legacy components were designed for SnPb soldering, making compatibility generally excellent. However, modern RoHS-compliant components may have pure tin or NiPdAu plating that behaves differently with SnPb.

Termination	Compatibility	Wetting	Concerns
SnPb Plated	■ Excellent	Excellent	Ideal match — legacy components
Pure Tin (Sn)	■ Good	Good	Tin whisker growth (long-term reliability)
Matte Sn	■ Good	Good	Lower whisker risk than bright Sn
NiPdAu	■ Excellent	Excellent	Gold embrittlement if > 3μm Au
SnBi	■■ Caution	Good	Low melting — reflow can remelt termination
SnAg	■ Good	Good	Standard RoHS termination
SnCu	■ Good	Good	Slightly slower wetting
OSP (Cu)	■ Good	Good	Requires active flux; short shelf life
Immersion Ag	■ Excellent	Excellent	Silver migration risk in high humidity
Immersion Sn	■ Good	Good	Whisker risk; IMC growth over time
ENIG (Ni/Au)	■ Excellent	Excellent	Black pad syndrome (brittle Ni-P layer)

Table 3: Component Termination Compatibility with SnPb Solder Paste

Key Compatibility Rules:

- **SnPb-plated terminations:** Perfect match — ideal for exempt applications
- **Pure Sn terminations:** Good wetting, but monitor for tin whiskers in high-stress applications
- **NiPdAu:** Excellent solderability, no whiskers, but expensive
- **OSP:** Cost-effective but requires paste with strong activators
- **ENIG:** Widely used but risk of 'black pad' — brittle fracture at Ni/solder interface
- **NEVER use SnPb paste on SnBi-plated components** — forms 96°C eutectic, catastrophic failure

6. STORAGE, HANDLING & SHELF LIFE

6.1 Refrigeration Requirements

Solder paste is a chemically active system. Elevated temperatures accelerate flux degradation, oxidation, and viscosity changes. Proper cold storage is critical even for SnPb paste.

- **Storage Temperature:** 0–10°C (32–50°F) — standard refrigerator, NOT freezer
- **Freezing Risk:** Below 0°C can separate flux from powder, ruining rheology
- **Shelf Life (Sealed):** 6 months at 0–10°C for Type III; 3 months for Type V/VI
- **Shelf Life (Opened):** 8–12 hours at room temperature after opening jar
- **Transport:** Use insulated cold packs; avoid direct sunlight exposure

6.2 Thawing Procedure

Improper thawing causes condensation (moisture ingress) and temperature shock. Follow this sequence:

1. Remove from refrigerator and place at **room temperature (20–25°C)**
2. Allow **minimum 4 hours** to reach ambient temperature naturally
3. **DO NOT force-thaw** with heat guns, ovens, or hot water baths
4. Wipe any external condensation from jar before opening
5. Record thaw time on jar label (traceability requirement)

6.3 Stencil Printing Life

Once on the stencil, solder paste begins to dry out due to solvent evaporation and flux activation. SnPb paste generally has **longer stencil life** than lead-free due to lower oxidation rates.

- **Stencil Life:** 8–12 hours at 20–25°C, 40–60% RH (longer than SAC305)
- **Response-to-Pause:** Paste must recover viscosity after 1-hour idle; test before resuming
- **Scrap paste** that exceeds stencil life — do NOT return to original jar
- **Mixing old + new paste** is prohibited — contaminates fresh batch

■ ■ **FIFO Rule:** First-In-First-Out inventory management is mandatory. Use oldest paste first. Expired paste must be discarded or returned to supplier for metal reclamation. Never use paste past its expiration date — wetting and voiding defects will increase.

7. REFLOW PROFILE FOR SNPB SOLDER PASTE

SnPb reflow is simpler and more forgiving than lead-free due to the lower melting point. The profile provides a wide process window with excellent wetting and minimal defects.

7.1 Standard Sn63Pb37 Reflow Profile

Zone	Temperature	Time / Rate	Purpose
Preheat	25 → 100°C	60–90 sec 1–3°C/sec	Evolve solvents, activate flux gently
Soak / Thermal Equalization	100 → 150°C	60–90 sec	Equalize temperature across board; complete flux activation
Ramp to Peak	150 → 210°C	1.5–3°C/sec	Rapid heating to liquidus; minimal thermal shock
Reflow (Above Liquidus)	> 183°C	30–60 sec	Solder melts, wets pads and leads, forms joints
Peak Temperature	210–230°C	10–20 sec at peak	Maximum wetting; IMC formation
Cooling	230 → 100°C	< 6°C/sec	Solidify joints; fine grain structure

Table 4: Recommended Sn63Pb37 Reflow Profile Parameters

7.2 Critical Profile Rules

- **Max ramp rate:** 3°C/sec to prevent component cracking and delamination
- **Min TAL:** 30 seconds above 183°C to ensure complete wetting
- **Max TAL:** 90 seconds to prevent excessive IMC growth and PCB degradation
- **Peak limit:** 240°C absolute maximum; 210–230°C typical
- **Cooling rate:** 2–6°C/sec optimal; too slow causes coarse grain structure

7.3 Comparison: SnPb vs SAC305 Reflow

Parameter	Sn63Pb37	SAC305	Advantage
Peak Temperature	210–230°C	245–250°C	SnPb — lower thermal stress
Energy Consumption	~30–40% less	Baseline	SnPb — lower utility costs
Process Window	Wide (40–50°C)	Narrow (20–30°C)	SnPb — more forgiving
Voiding	Low (< 10%)	Moderate (15–25%)	SnPb — better X-ray yields
Wetting Speed	Excellent	Good	SnPb — faster production
Joint Appearance	Bright, shiny	Dull, grainy	SnPb — easier inspection
Tombstoning	Rare	Common	SnPb — fewer defects

Tip Life (Hand)	2–3× longer	Baseline	SnPb — lower consumable cost
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Table 5: SnPb vs SAC305 Reflow Process Comparison

8. QUALITY CONTROL & COMMON DEFECTS

8.1 Visual Inspection (IPC-A-610)

SnPb joints have a **bright, shiny, smooth** appearance that is easy to inspect visually. IPC-A-610 acceptance criteria for SnPb:

- **Acceptable:** Bright, smooth fillet with concave meniscus
- **Acceptable:** Slight graininess on large joints (normal)
- **Reject:** Dull, grainy, or frosted appearance (overheating or contamination)
- **Reject:** Cracks, voids, or insufficient wetting
- **Reject:** Excessive solder (solder balls, bridging)

8.2 Common SnPb Paste Defects

Cold Joint

Dull, rough surface with visible grain boundaries. Caused by insufficient heat, movement during cooling, or contaminated surfaces. In SnPb, cold joints are easier to identify due to the loss of characteristic shine.

→ **Fix:** Increase peak temperature or TAL; verify thermocouple placement; check flux activity

Bridging

Excessive solder forms a bridge between adjacent pins or pads. Common in fine-pitch QFP and QFN packages. Caused by excessive paste volume, poor stencil design, or incorrect reflow profile.

→ **Fix:** Reduce stencil aperture, optimize reflow ramp rate, verify board support

Insufficient Solder

Fillet height below 25% of lead thickness or pad not fully covered. Caused by insufficient paste deposition, poor wetting, or paste dried out on stencil.

→ **Fix:** Check stencil aperture ratio, verify paste viscosity, replace old paste

Solder Balling

Small solder spheres scattered around the joint. Less common in SnPb than SAC305 but can occur from rapid solvent evolution or paste contamination.

→ **Fix:** Extend preheat soak, reduce ramp rate, verify paste storage conditions

Lifted Pad

PCB pad delaminates from the substrate. Usually a PCB quality issue (insufficient copper adhesion) rather than a solder problem, but excessive heat can exacerbate it.

→ **Fix:** Verify PCB supplier quality; reduce peak temperature; improve board support

Tin Pest (Allotropic Transformation)

Rare phenomenon where high-tin alloys ($\text{Sn} > 60\%$) transform to brittle α -tin at temperatures below 13°C . Causes solder joints to crumble. Extremely rare in normal operating environments.

→ **Fix:** Use Sn60Pb40 or lower tin content for extreme cold environments

8.3 Intermetallic Compound (IMC) Analysis

SnPb forms **Cu₆Sn₅** and **Cu₃Sn** intermetallic layers at the solder-copper interface. Target IMC thickness: **1–3µm**. SnPb generally forms thinner, more stable IMC layers than SAC305, contributing to long-term reliability.

- **Thin IMC (< 1µm):** Indicates insufficient heat or time — weak bond
- **Optimal IMC (1–3µm):** Strong, reliable joint
- **Thick IMC (> 5µm):** Brittle layer — prone to fracture under stress

9. SAFETY, HANDLING & PROCESS SEPARATION

9.1 Health Hazards

Lead is a neurotoxin. Proper handling procedures are mandatory in any facility using SnPb paste:

- **Wash hands** thoroughly after handling solder paste
- **Do NOT eat, drink, or smoke** in soldering work areas
- **Use fume extraction** — solder fumes contain lead oxide and flux vapors
- **Wear gloves** when handling bare paste or cleaning stencils
- **Eye protection** required during stencil cleaning and paste mixing
- **Pregnant women** should not handle leaded solder materials

9.2 Waste Disposal

- **Scrap paste:** Collect in labeled hazardous waste containers
- **Stencil wipes:** Disposed as lead-contaminated waste
- **Squeegees and tools:** Cleaned in designated lead-only stations
- **Never dispose** of leaded solder paste in general trash or drains
- **Expired paste:** Return to supplier for certified metal reclamation

9.3 Workplace Separation (Critical)

If both leaded and lead-free lines exist in the same facility, implement **strict physical separation**:

- **Physical separation:** Different rooms or clearly marked zones
- **Color coding:** RED for leaded, GREEN for lead-free
- **Dedicated tools:** Stencils, squeegees, spatulas, and mixers must not mix
- **Dedicated ovens:** Separate reflow ovens preferred; if shared, thorough purge required
- **Labeling:** All paste jars, stencils, and WIP boards clearly marked
- **Training:** All operators trained on cross-contamination risks
- **Quality audit:** Periodic XRF testing of random joints to verify alloy composition

■ **Cross-Contamination Risk:** Mixing even small amounts of SnPb into a lead-free process voids RoHS compliance and can cause catastrophic joint failures (especially with SnBi alloys). Always verify alloy type before loading any paste into equipment. Implement barcode scanning for paste jars to prevent human error.

10. SOLDER PASTE SELECTION DECISION MATRIX

Use this matrix to select the optimal leaded solder paste for your specific exempt application.

Application	Recommended Alloy	Powder Type	Flux	Notes
General Exempt SMT	Sn63Pb37	Type III	RMA No-Clean	Workhorse choice
Medical Devices	Sn63Pb37	Type III	M1 Water-Wash	Zero residue mandatory
Military / Aerospace	Sn62Pb36Ag2	Type III	RMA / M1 Wash	Ag-plated terminations common
RF / Microwave	Sn62Pb36Ag2	Type III	RMA No-Clean	Silver traces on PCBs
High-Volume Legacy	Sn60Pb40	Type III	RMA No-Clean	Cost-optimized pre-RoHS
Step-Soldering (2nd Pass)	Sn50Pb50	Type III	RMA No-Clean	Higher melt range prevents remelt
Repair / Rework	Sn63Pb37	Type III	RMA Wire Solder	Match original alloy when possible
Fine Pitch QFP	Sn63Pb37	Type IV	RMA No-Clean	Smaller powder for better release
0201 / 01005 Passives	Sn63Pb37	Type V	RMA No-Clean	Ultra-fine stencil or jet print
Power Modules	Sn62Pb36Ag2	Type III	RMA No-Clean	Ag content improves creep resistance

Table 6: Application-Based Leaded Solder Paste Selection Matrix

11. QUICK REFERENCE SUMMARY

Alloy Selection:

- **Sn63Pb37** — 90% of exempt applications; eutectic, best wetting, lowest cost
- **Sn62Pb36Ag2** — Silver-plated terminations (military, RF, aerospace)
- **Sn60Pb40** — Legacy cost-driven production, slightly more ductile
- **Sn50Pb50** — Step-soldering, maximum ductility, wider melt range

Powder Type:

- **Type III** — Standard SMT (0402, 0603, SOIC, QFP)
- **Type IV** — Fine pitch (0201, 0.5mm QFP, small QFN)
- **Type V/VI** — Ultra-fine (01005, jet dispensing, SiP)

Flux Selection:

- **RMA No-Clean** — Traditional standard for SnPb electronics
- **M1 Water-Wash** — High reliability; mandatory cleaning step
- **L0 No-Clean** — Ultra-low residue for sensitive impedance circuits
- **RA** — **NEVER for PCBA** — Corrosive, reserved for plumbing only

Storage Rules:

- **0–10°C** refrigerated, sealed, FIFO inventory
- **4-hour minimum thaw** at room temperature before use
- **8–12 hour stencil life** at room temperature (longer than lead-free)
- **Never return** used paste to original container

Reflow Essentials:

- **Peak:** 210–230°C for Sn63Pb37
- **TAL:** 30–60 seconds above 183°C
- **Ramp:** 1.5–3°C/sec max
- **Cooling:** 2–6°C/sec optimal

Safety & Separation:

- **RED color code** for all leaded tools, stencils, and zones
 - **Physical separation** from lead-free lines mandatory
 - **XRF audit** random joints to verify no cross-contamination
 - **Lead is a neurotoxin** — fume extraction and hand washing mandatory
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References & Standards:

- IPC/J-STD-005: Requirements for Soldering Pastes
- IPC/J-STD-006: Requirements for Electronic Grade Solder Alloys
- IPC/J-STD-004: Requirements for Soldering Fluxes
- IPC-A-610: Acceptability of Electronic Assemblies
- RoHS Directive 2011/65/EU Exemptions List
- MIL-STD-883: Test Methods and Procedures for Microelectronics
- MIL-STD-202: Environmental Test Methods
- Kester Solder Technical Handbook

End of Document — Leaded Solder Paste Complete Technical Guide